CATEGORICAL MODULI SPACES VS CATEGORICAL QUOTIENTS UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3246-3252 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3254-3274, the complete author proof (19 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 42 of queue_stacks-stacks-more-morphisms.json (pb=1036, stmt 3246-3252, proof 3254-3274).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04WY: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3274–3274. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06FH: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3274–3274. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 048J: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3274–3274. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-quotient-compare: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3217–3224. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *With assumption and notation as in Lemma lemma-quotient-compare. Then f is a (uniform) categorical moduli space if and only if ϕ is a (uniform) categorical quotient. Similarly for moduli spaces in a full subcategory.*

Proof. It is immediate from the 1-to-1 correspondence established in Lemma lemma-quotient-compare that f is a categorical moduli space if and only if ϕ is a categorical quotient (Quotients of Groupoids, Definition 048J). If $Y' \to Y$ is a morphism, then $U' = Y' \times_Y U \to Y' \times_Y \mathcal{X} = \mathcal{X}'$ is a surjective, flat, locally finitely presented morphism as a base change of $U \to \mathcal{X}$ (Criteria for Representability, Lemma 06FH). And $R' = Y' \times_Y R$ is equal to $U' \times_{\mathcal{X}'} U'$ by transitivity of fibre products. Hence $\mathcal{X}' = [U'/R']$, see Algebraic Stacks, Remark 04WY. Thus the base change of our situation to Y' is another situation as in the statement of the lemma. From this it immediately follows that f is a uniform categorical moduli space if and only if ϕ is a uniform categorical quotient. □